

An Augmented Reality Approach for Phantom Limb Pain: A Case Report

Objective: This case report introduces a new augmented reality mirror therapy program to manage phantom limb pain (PLP).

Methods: Our augmented reality mirror therapy program for individuals with unilateral limb loss uses an infrared camera to capture real time images of the intact limb and uses custom designed software to create a digital mirror limb (phantom limb) image. Then, the user can observe their intact limb and phantom limb on a monitor with the ability to control their point of view in 360 degrees of freedom. We completed one 30 minute session using this program with a seventy year-old male who underwent a left transtibial amputation due to Kaposi's sarcoma with severe, persistent PLP, described as sharp, electrical impulses throughout his phantom limb. The patient was encouraged "freestyle" movement with his phantom limb and instructed in traditional mirror therapy exercises. Prior to this program, the patient's treatment paradigm included medications, physical therapy, meditation, visualization, physical modalities and traditional mirror therapy without resolution of PLP.

Results: Preliminary qualitative findings revealed decreased pain scores in his phantom limb. He stated, "PLP is alleviated when looking straight down at my knees." The patient emphasized a strong response from real time control of his point of view. When shown an oblique view of his phantom limb he stated, "This is much more convincing... this looks like a real person. It hasn't made the pain go away but it's become more comfortable."

Discussion: This case report reveals the potential impact of a new augmented reality mirror therapy program. This new program retains all of the key features of traditional mirror therapy, including portability, while improving upon point of view control. Further development of this design and study of its impact is warranted.